

Advanced Techniques for Cryptocurrency Funds Lost Prediction

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I. Introduction

Cryptocurrency is a form of digital or virtual currency that uses cryptography for secure financial transactions, eliminating the need for centralized authorities like banks. Cryptocurrencies operate on blockchain technology, which is a decentralized ledger that records all transactions across a network of computers. Popular examples include Bitcoin, Ethereum, and Ripple. Transactions in cryptocurrencies are made by transferring digital assets from one user's digital wallet to another's. These transactions are verified by network nodes through cryptographic methods and recorded on the blockchain, ensuring transparency and immutability.

Despite its advantages, cryptocurrency is highly susceptible to fraud due to its decentralized and pseudonymous nature. Fraudulent activities include hacking of wallets and exchanges, phishing attacks, Ponzi schemes, rug pulls, and initial coin offering (ICO) scams. Additionally, the lack of stringent regulations in some jurisdictions provides opportunities for malicious actors to exploit the system. The anonymity of cryptocurrency transactions further complicates the detection and prevention of fraudulent activities, making the ecosystem a target for cybercriminals.

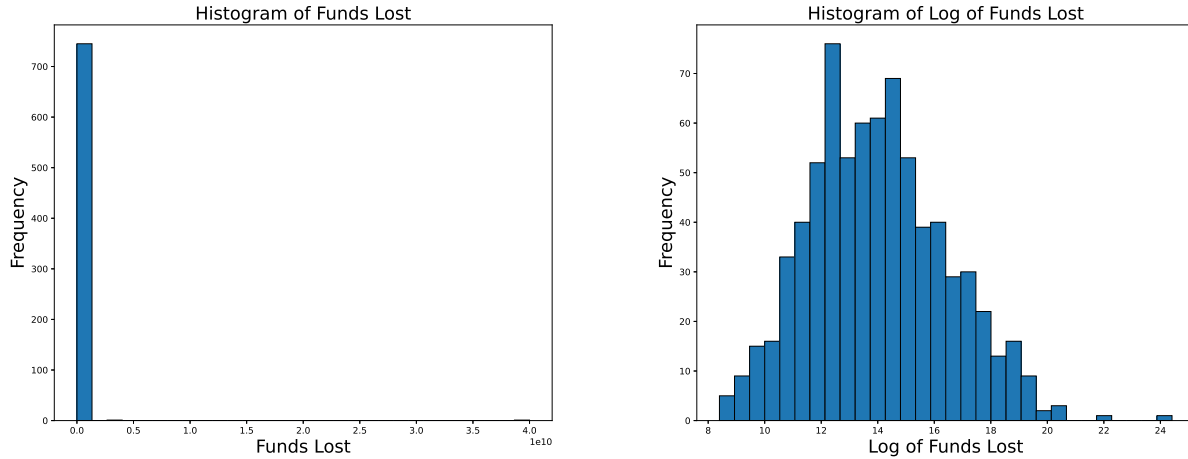
Studying the size of cryptocurrency frauds, particularly the funds lost, is crucial for several reasons. First, it helps quantify the economic impact of fraudulent activities on individual investors, organizations, and the broader financial ecosystem. Second, understanding the scale of fraud can guide regulatory bodies in developing robust policies to secure digital assets. Finally, insights into fraudulent patterns and methods can aid in developing technological solutions to improve the security and trustworthiness of cryptocurrency systems, fostering wider adoption and ensuring financial stability in this emerging market. The objective of this study is to build a model to predict fund lost to fraudulent cryptocurrency activities using advanced regression and machine learning techniques.

II. Data

The dataset for this study was downloaded from the internet. It has five variables, the funds lost which happens to be the target variable for this study and four other variables: Date which is the

date on which the fraud occurred, the Category, the Issue which is the specific fraud, and the Chain. Below is the histogram of the Lost Amount in dollars.

Figure 1. Lost Funds Distribution



From the histogram of Lost Funds, we see that most of the amounts are less than a million Dollar which is why there is a huge bar to the left. To make the target distribution slightly normal, we apply the log transformation which gives us the distribution on the right which appears to be a bit more normal. This will help in our subsequent predictive modeling. After the log transformation, the average fund lost was 14.00 with a standard deviation of about 2.42. This shows that there is not a huge variability in the target. I also transformed the date to the number of days since 01/01/2011. The average for that is 3958.25. This shows that the average year that the fraud cases occurred was between 2022 and 2023. There are 10 Categories, 6 Issues and 5 Chains. The proportions for each class within each of the above categorical variables is presented in Table 1 below.

Table 1—Summary Statistics

	Mean	SDV	Minimum	Maximum
Funds Lost	13.999	2.419	8.398	24.412
Days_Since_2011	3958.253	964.037	163.000	4973.000
Category_Borrowing	0.111	0.314	False	True
Category_Bridge	0.035	0.183	False	True
Category_CeFi	0.143	0.351	False	True
Category_Exchange	0.143	0.351	False	True
Category_GamingM	0.039	0.193	False	True
Category_NFT	0.055	0.228	False	True
Category_Other	0.234	0.424	False	True
Category_Stablecoin	0.025	0.158	False	True
Category-Token	0.119	0.324	False	True
Category_YieldAggr	0.095	0.293	False	True
Issue_AccessControl	0.252	0.434	False	True
Issue_FlashLoanAttack	0.159	0.366	False	True
Issue_OracleIssue	0.023	0.149	False	True
Issue_Other	0.486	0.500	False	True
Issue_Phishing	0.051	0.220	False	True
Issue_Reentrancy	0.029	0.169	False	True
Chain_ARBI	0.041	0.200	False	True
Chain_BSC	0.246	0.431	False	True
Chain_CEX	0.142	0.349	False	True
Chain_ETH	0.399	0.490	False	True
Chain_Other	0.171	0.377	False	True

III. Methodology

To effectively model Funds Lost using the dataset, I propose to explore a range of advanced machine learning methodologies, specifically focusing on methods that can capture nonlinear relationships and interactions between variables.

1) Linear Models

In this study, we employ a Generalized Linear Model (GLM) as a baseline model for predicting Fund Lost. GLM is an extension of the ordinary linear regression model that allows the dependent variable (or response) to follow distributions other than the normal distribution. GLMs are widely used in statistical modeling to handle cases where the assumptions of linear regression (e.g., normality and constant variance) do not hold.

2) Tree-Based Models

Tree-based models are powerful for predictive tasks because they handle nonlinear relationships well and can manage high-dimensional data. In this project, we propose to use the following tree-based methods:

- a) Random Forests: Random Forests are an ensemble learning method that constructs multiple decision trees and aggregates their results to improve model robustness and reduce overfitting (Segal, 2004). This approach is highly interpretable, as we can assess the importance of each variable in predicting Funds Lost, helping us understand the drivers of this in the dataset. Bagging is a special case of Random Forest where all predictors are considered at each split (Breiman, 1996).
- b) Gradient Boosting Machines (GBM): GBMs iteratively build decision trees to correct errors made by previous models, allowing for fine-tuning and greater accuracy. Variants of GBMs, such as XGBoost and LightGBM, are computationally efficient and often produce state-of-the-art results for structured data tasks (Ke et al., 2017). These models allow for tuning of learning rate, tree depth, and number of estimators, providing flexibility in capturing complex relationships in the data.
- c) Bayesian Additive Regression Trees (BART): BART provide a probabilistic approach to regression, combining the flexibility of tree-based models with Bayesian inference. BART models predict the response variable by summing the contributions of multiple small trees, each representing a different feature interaction. According to Chipman et al. (2010), this approach allows for a robust prediction interval, giving insight into the

uncertainty of predictions—an advantage over traditional tree-based models. BART is especially suitable for this dataset because it can model complex, non-linear interactions without extensive feature engineering. Moreover, its Bayesian nature allows it to account for uncertainty in predictions, which is valuable for financial forecasting, as it enables stakeholders to evaluate risk alongside the predicted prices.

Tree-based models are advantageous for this dataset due to their ability to automatically handle categorical features which are common in real-world data. Additionally, they are resilient to multicollinearity, which is often present in high-dimensional data.

3) Generalized Additive Models (GAM)

Generalized Additive Models (GAM) are an extension of traditional linear models that allow for non-linear relationships between the response and predictor variables by applying smooth functions to each variable (Hastie, 2017). GAMs provide a balance between interpretability and flexibility, as they maintain additive structure while allowing individual predictors to have non-linear effects. This can be beneficial for house price prediction, where certain features (e.g., year built, lot size) may have a non-linear influence on price. GAMs are advantageous for interpretability, as they clearly show the effect of each feature on the response variable, which is crucial for explaining predictions to stakeholders.

IV. Results

A. Linear Models

The linear regression model output provides insights into the relationship between the independent variables and the Funds Lost. The model is based on 597 observations in the training set. The R^2 value is 0.239, indicating that approximately 23.9% of the variance in “Funds Lost” is explained by the model. The adjusted R^2 (0.214) is slightly lower, accounting for the number of predictors in the model. The F-statistic is significant at $p < 0.01$, suggesting that the model as a whole is statistically significant.

One significant feature is Category_Token ($\beta = -2.330^{***}$, $p < 0.01$). Tokens are associated with a significant reduction in Funds Lost compared to the Borrowing. This negative coefficient suggests that incidents involving tokens generally result in lower losses compared to borrowing.

Table 2—Linear Regression

	Dependent variable: Funds Lost
	(1)
Category_Bridge	1.185** (0.565)
Category_CeFi	0.523 (0.580)
Category_Exchange	-0.379 (0.371)
Category_GamingM	-0.954* (0.563)
Category_NFT	-1.992*** (0.483)
Category_Other	-0.624* (0.349)
Category_Stablecoin	1.810*** (0.652)
Category_Token	-2.330*** (0.402)
Category_YieldAggr	-0.502 (0.416)
Chain_BSC	0.323 (0.487)
Chain_CEX	1.455* (0.754)
Chain_ETH	1.087** (0.472)
Chain_Other	0.925* (0.502)
Days_Since_2011	0.001*** (0.000)
Issue_FlashLoanAttack	-0.636** (0.324)
Issue_OracleIssue	0.309 (0.639)
Issue_Other	-0.565** (0.237)
Issue_Phishing	-0.704 (0.467)
Issue_Reentrancy	-1.258** (0.589)
intercept	12.063*** (0.831)
Observations	597
R^2	0.239
Adjusted R^2	0.214
Residual Std. Error	2.174 (df=577)
F Statistic	9.546*** (df=19; 577)

Note:

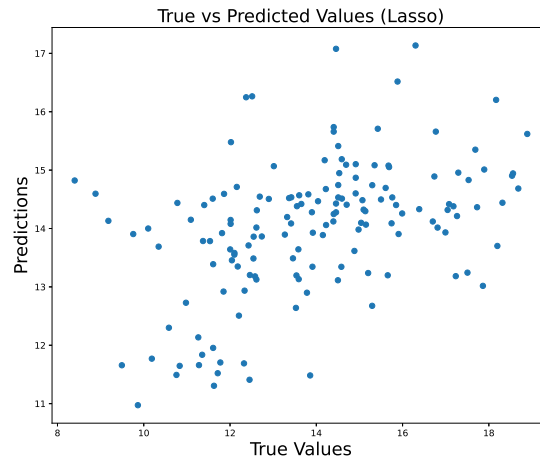
*p<0.1; **p<0.05; ***p<0.01

Also, another important feature is `Category_Stablecoin` ($\beta = 1.810^{**}$, $p < 0.01$). Stablecoin-related incidents are positively and significantly associated with funds lost compared to Borrowing. This implies that breaches involving stablecoins tend to lead to higher financial losses.

The variable `Chain_CEX` ($\beta = 1.455^*$, $p < 0.05$) also stands out. Centralized exchange (CEX)-related chains show a positive relationship with Funds Lost. This could indicate that vulnerabilities in centralized exchanges lead to significant losses compared to ARBI. For `Issue_Reentrancy` ($\beta = -1.258^*$, $p < 0.05$), Re-entrance issues are associated with a significant reduction in Funds Lost compared to the Access Control issues. While still serious, re-entrance issues seem to result in comparatively smaller losses compared to access control issues. Finally, the variable `Days_Since_2011` ($\beta = 0.001^{***}$, $p < 0.01$) is small but statistically significant. It suggests that as time progresses, there is a slight increase in the funds lost in incidents, possibly reflecting the growing scale of transactions or vulnerabilities in the blockchain ecosystem.

In all, the findings suggest that stable coins, centralized exchanges, and temporal trends are critical factors in understanding the magnitude of financial losses. This is also consistent with the result of the GLM model which is presented in Table 4 in the Appendix.

Figure 2. Lasso Scatter Plot

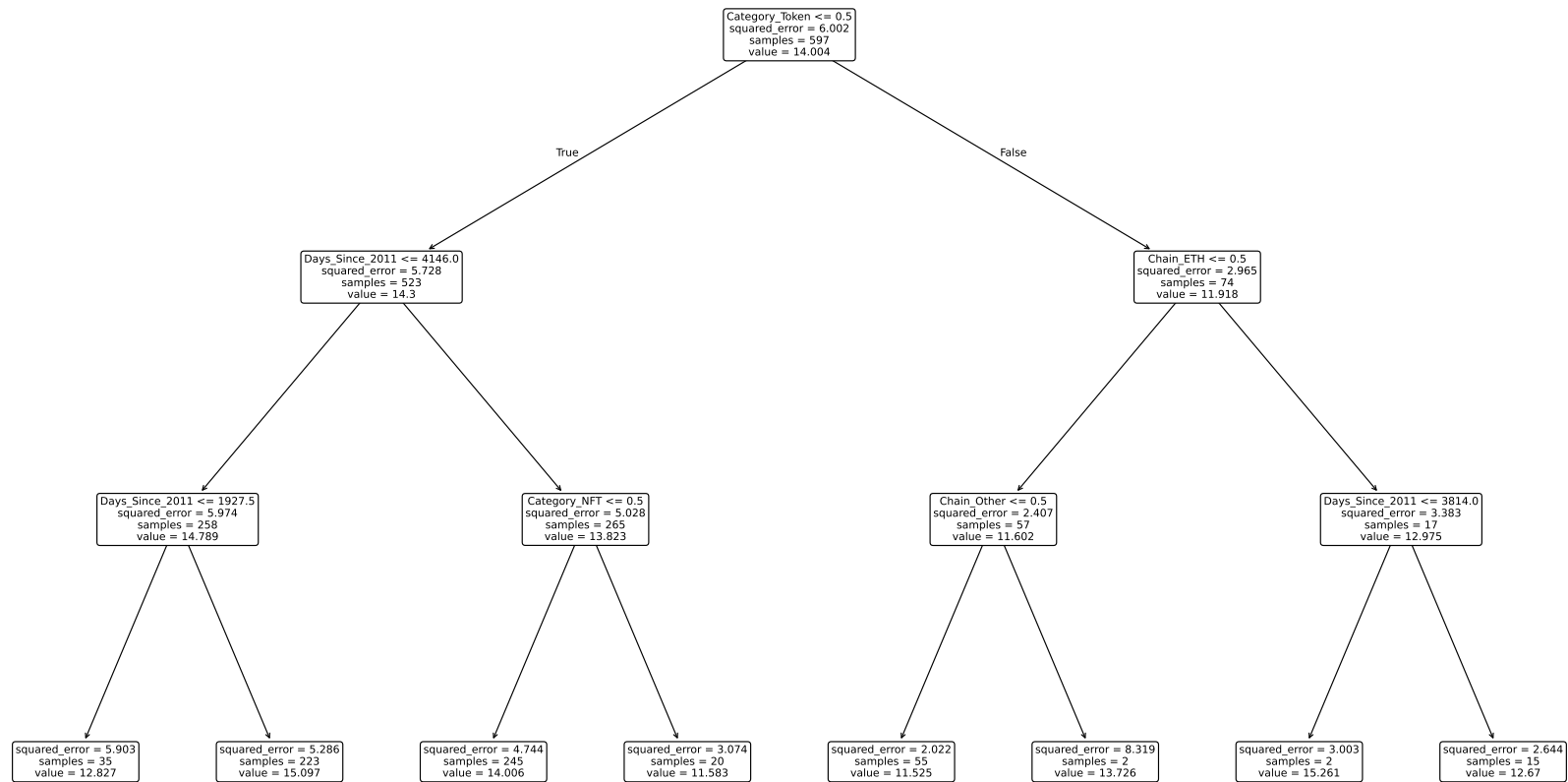


From the linear regression, we could clearly see that most features were not statistically significant. To address this, I employed Lasso which is a regularization method. This was intended to help do feature selection. The graph of actual vs predicted values in the test set is shown in Figure 2 above. We see that the Lasso models performs fairly well on the test set but not as well which means that there is the need to other non-linear approaches.

B. Tree-based Models

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Figure 3. Regression Tree (Optimal Depth)



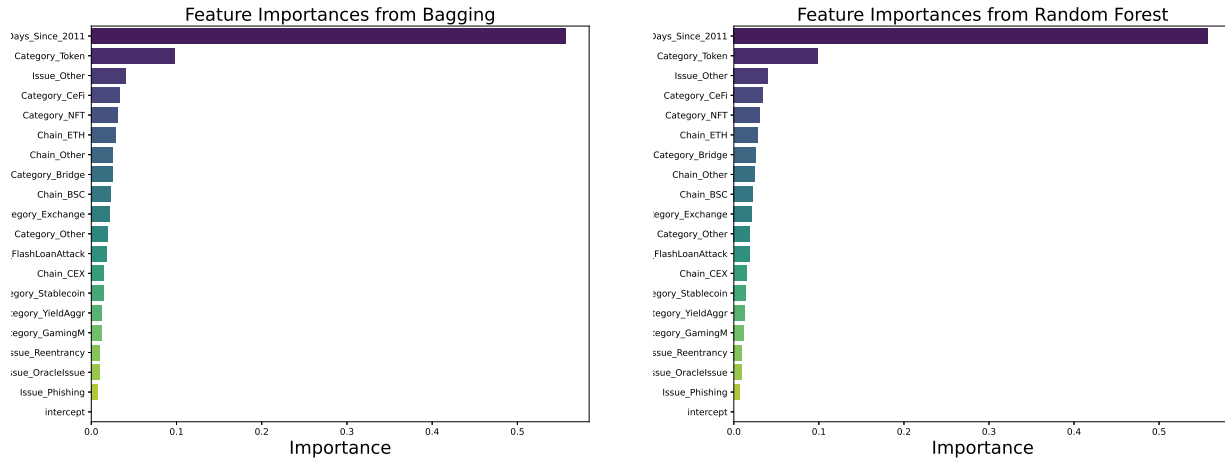
The regression tree provides a structured, hierarchical view of the factors influencing Funds Lost, with the target variable being progressively split at each node. The root node splits based on `Category_Token` (≤ 0.5), indicating that the presence or absence of token-related incidents is the primary factor in predicting Funds Lost. The mean value of Funds Lost at the root node is 14.004, with a squared error of 6.002.

For the leftmost terminal node, the natural log of Funds Lost is predicted to be 12.827 if the Category is not Token and the `Days_Since_2011` ≤ 1927.5 . This means that categories other than Taken are associated with lower Funds Lost. The natural log of Funds Lost is predicted to be 15.097 if the Category is not Token and `Days_Since_2011` is between 1927.5 and 4146. Also, natural log of Funds Lost is predicted to be 14.006 if the Category is neither Token nor NFT and `Days_Since_2011` ≤ 4146 . This is the group where most of the incidence in the training set is reported. If the Category is NFT and `Days_Since_2011` ≤ 4146 , the predicted natural log of Funds Lost is 11.583.

The natural log of Funds Lost is predicted to be 11.525 if the Category is Token and Chain is neither ETH or Others. But if Category is Token and Chain is Others, then the natural log of Funds Lost is predicted to be 13.726. When Category is Token, Chain is ETH and `Days_Since_2011` ≤ 3814 , natural log of Funds Lost is predicted to be 15.261. This group contains the highest average Funds Lost. Finally, if Category is Token, Chain is ETH and `Days_Since_2011` > 3814 , natural log of Funds Lost is predicted to be 12.67.

One of the critiques of regressions trees is that is is mostly non-robust and is very sensitive to the training set. To mitigate this flaw, I looked at some ensemble tree method. To assess the significance of features in the ensemble tree, we use the variable (feature) importance plot. Both the baggibg and random forest models indicate that `Days_Since_2011` is the most significant predictor, demonstrating its dominant role in explaining the variance in the target variable. This suggests that temporal factors, such as the time since 2011, heavily influence the magnitude of funds lost in incidents. The second most important variable is `Category_Token`, which consistently ranks high in importance across both models. This reflects the critical role that token-related categories play in predicting financial losses, likely due to their distinct characteristics in terms of vulnerabilities or incident frequency. `Issue_Other` is another significant variable, highlighting that issues grouped in this category contribute substantially to variations in the outcome. This indicates the need to investigate these issues further to understand their specific impacts. Other features such as `Category_CeFi`, `Category_NFT`, and `Chain_ETH` show moderate importance in

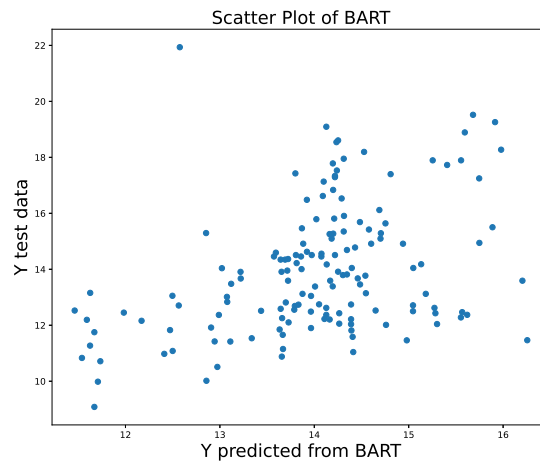
Figure 4. Feature Importance Plots



both models. These features suggest that incidents related to centralized finance, non-fungible tokens, and Ethereum-based chains also play noteworthy roles in predicting Funds Lost.

The similarity in the feature importance rankings between Bagging and Random Forest models reflects the robustness of these findings. However, minor differences in the relative importance of less significant features, such as Category_Stablecoin or Issue_Phishing, suggest slight variations in how the models interpret these predictors.

Figure 5. BART Scatter Plot



Another ensemble tree that I considered for this analysis is the BART. Figure 5 provides the plot for the predicted versus actually values from the test set. We observe that the BART models performs slightly better on the test data compared to the Lasso models we saw previously but still does not perfectly fit the test set.

C. Generalized Additive Models (GAM)

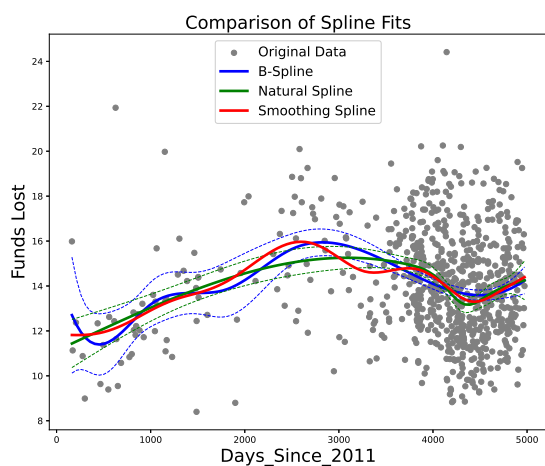
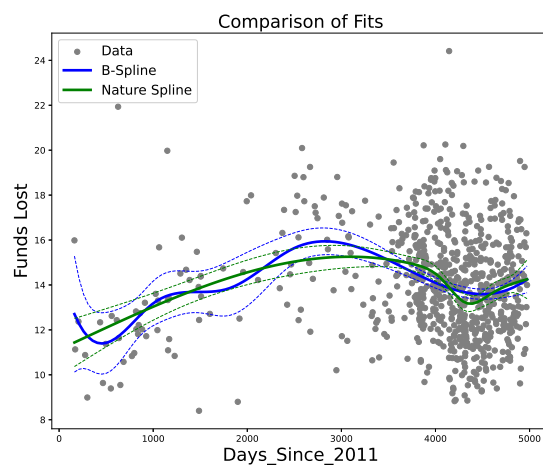
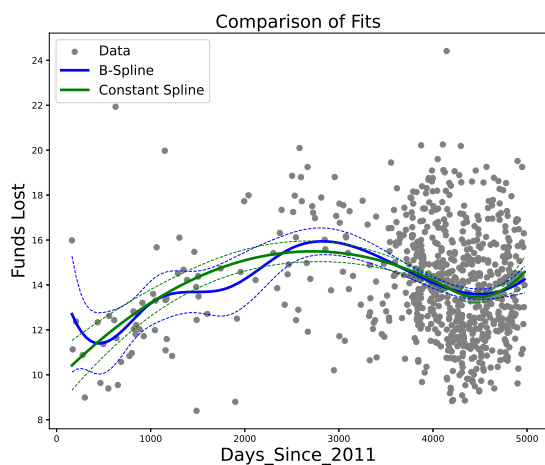
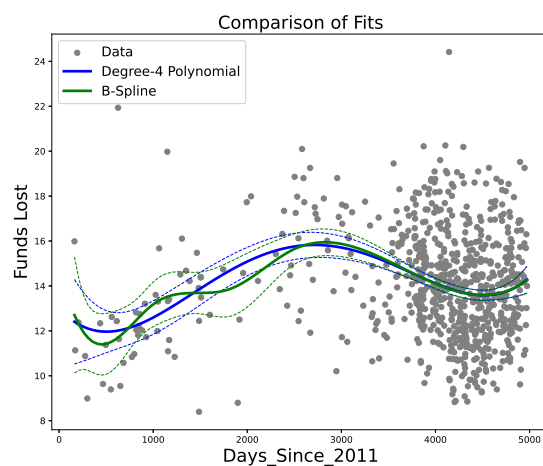
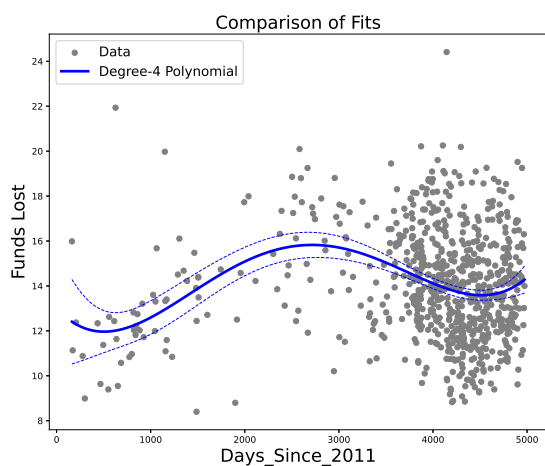
To further explore the model that fits the data best, we looked at the GAM. Figure 6 presents a comparison of different Generalized Additive Model (GAM) fits to predict Funds Lost as a function of Days_Since_2011. Various smoothing techniques, including degree-4 polynomial fits, B-splines, constant splines, natural splines, and LOWESS smoothing, are employed to evaluate their effectiveness in capturing the relationship between the variables.

The first panel demonstrates the fit using a degree-4 polynomial, which provides a reasonable approximation of the data but exhibits noticeable oscillations in regions with sparse data points. This overfitting tendency can introduce instability, particularly when extrapolating to unseen data. The second panel compares the degree-4 polynomial with a B-spline fit. The B-spline provides a smoother curve, effectively balancing flexibility with the prevention of overfitting. This results in a fit that captures the underlying trends of the data without being overly influenced by noise.

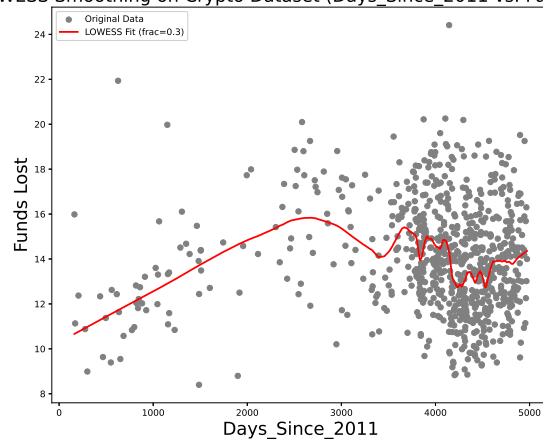
The third panel introduces a constant spline, which fits the data using piecewise constant segments. While this method is straightforward, it lacks the ability to model gradual transitions or more complex nonlinear trends, limiting its interpretability and accuracy for this dataset. The fourth panel evaluates the performance of B-splines versus natural splines. Both approaches yield similar results, but natural splines provide an additional constraint at the boundaries, leading to slightly improved stability in those regions. This highlights their suitability for capturing nonlinear trends in datasets with temporal boundaries.

The fifth panel compares the fits of B-splines, natural splines, and smoothing splines. Among these, the smoothing spline offers the highest degree of flexibility, allowing it to closely follow the observed data points. However, this flexibility may lead to overfitting in regions with high noise, necessitating careful parameter tuning. The final panel employs LOWESS (Locally Weighted Scatterplot Smoothing) to fit the data. LOWESS provides a highly flexible, localized fit, effectively capturing the nonlinear relationships in the data. The resulting smooth curve is particularly useful for visualizing trends without imposing a global functional form. Among the presented methods, B-splines and natural splines emerge as robust choices for modeling nonlinear trends, while LOWESS excels in exploratory data analysis due to its adaptability to local patterns.

Figure 6. Comparison of Different GAM Fits



LOWESS Smoothing on Crypto Dataset (Days_Since_2011 vs. Funds Los

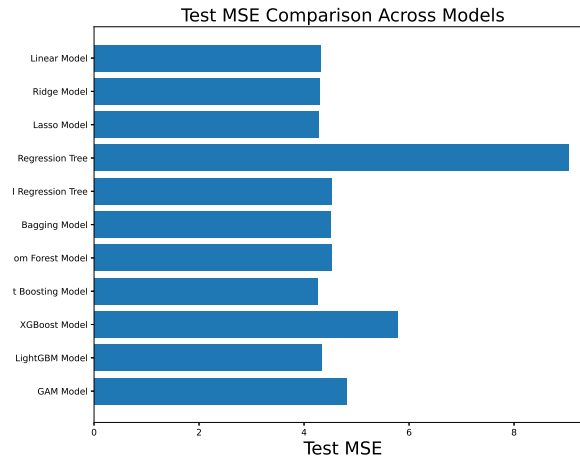


V. Conclusion

Table 3—Test MSE for Different Models

Model	Test MSE
Linear Model	4.314
Ridge Model	4.286
Lasso Model	4.278
Regression Tree	9.031
Pruned Regression Tree	4.516
Bagging Model	4.503
Random Forest Model	4.524
Gradient Boosting Model	4.253
XGBoost Model	5.772
LightGBM Model	4.339
GAM Model	4.801

Figure 7. Test MSE for Different Models



Based on the test MSE values, the Gradient Boosting Model achieves the lowest test MSE (4.253), making it the best-performing model for predicting “Funds Lost” among the options evaluated. It offers a good balance between accuracy and complexity, outperforming simpler models like Linear Regression, Ridge, and Lasso, as well as more complex ensemble methods like Bagging and Random Forest.

References

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VI. Appendix

Figure 8. Regression Tree

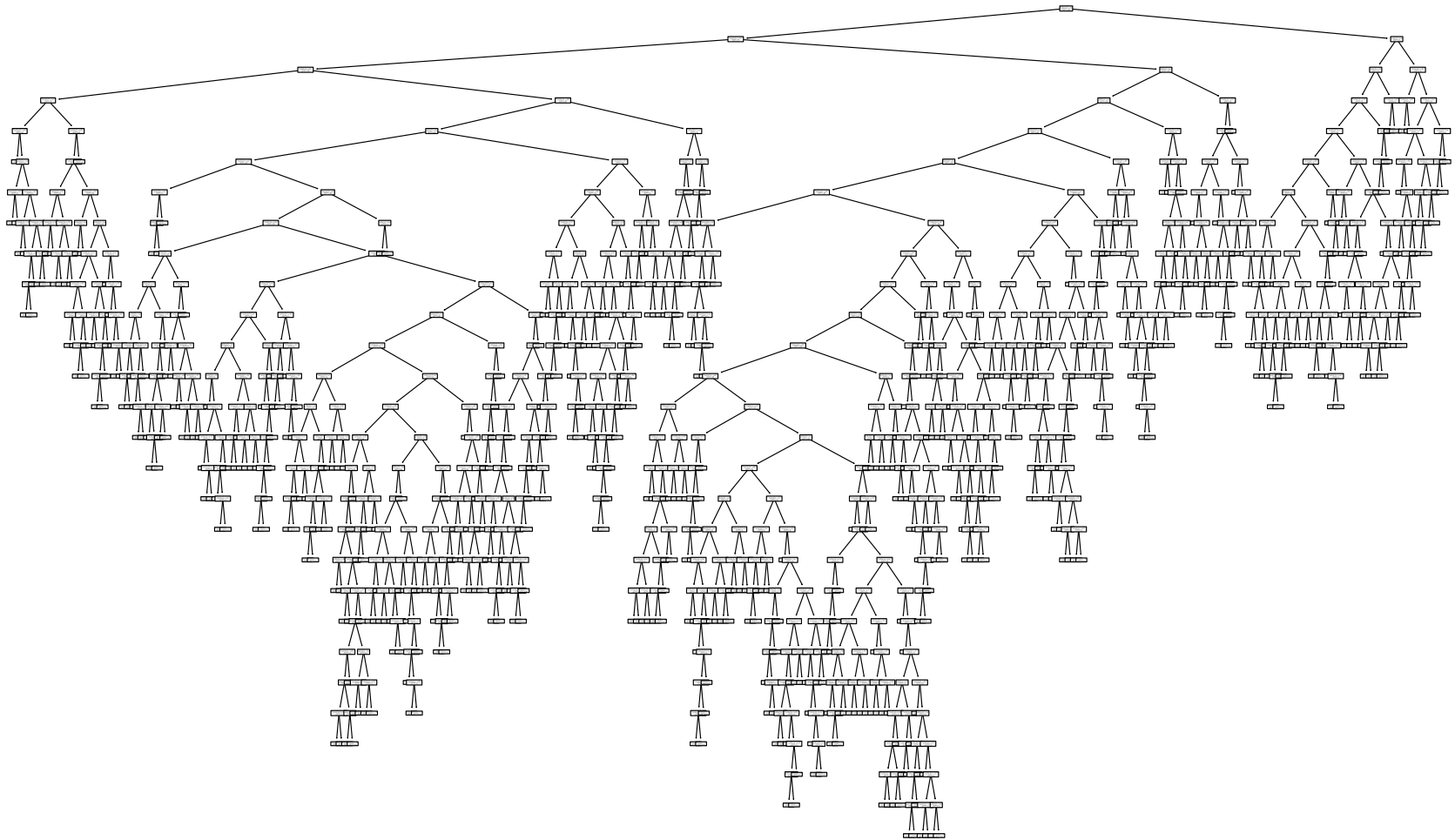


Table 4—GLM Regression

	Dependent variable: Funds Lost
	(1)
Category_Bridge	0.077* (0.040)
Category_CeFi	0.033 (0.041)
Category_Exchange	-0.026 (0.026)
Category_GamingM	-0.070* (0.040)
Category_NFT	-0.147*** (0.034)
Category_Other	-0.045* (0.025)
Category_Stablecoin	0.118** (0.046)
Category_Token	-0.179*** (0.029)
Category_YieldAggr	-0.034 (0.030)
Chain_BSC	0.022 (0.035)
Chain_CEX	0.100* (0.054)
Chain_ETH	0.078** (0.034)
Chain_Other	0.064* (0.036)
Days_Since_2011	0.000*** (0.000)
Issue_FlashLoanAttack	-0.045* (0.023)
Issue_OracleIssue	0.020 (0.045)
Issue_Other	-0.039** (0.017)
Issue_Phishing	-0.049 (0.033)
Issue_Reentrancy	-0.090** (0.042)
intercept	2.505*** (0.059)
Observations	597

Note:

*p<0.1; **p<0.05; ***p<0.01

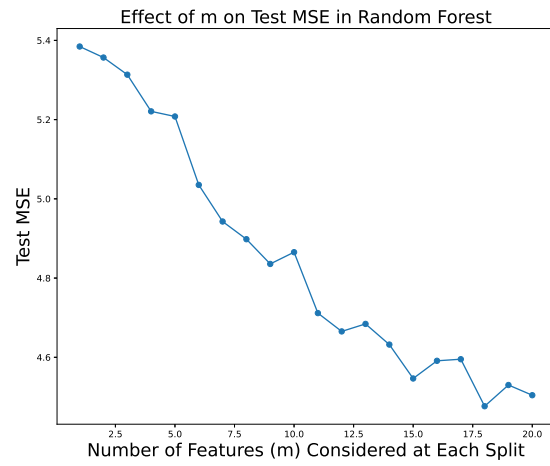
Figure 9. Effect of m on Test MSE in Random Forest

Figure 10. Exploratory Data Analysis

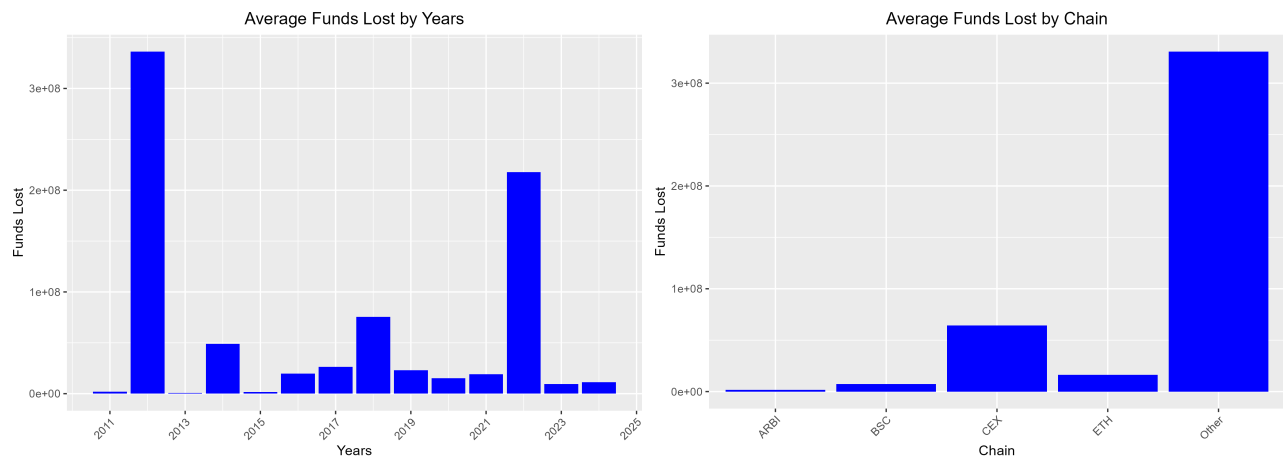


Figure 11. Exploratory Data Analysis

